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SEMICONDUCTOR STRUCTURE INCLUDING MEMORY SIGNAL TRANSMISSION LINE IN BACKSIDE REDISTRIBUTION

Abstract

A semiconductor structure includes a substrate, a data storage element, and first and second bit line sets. The substrate has a front side and a backside and includes a substrate layer, a first active layer, and a second active layer. The second active layer is proximal to the front side. The substrate layer is proximal to the backside. The data storage element extends across the first and second active layers. The first bit line set includes a first bit line and a first complementary bit line. At least one of the first bit line and the first complementary bit line is disposed on the front side of the substrate. The second bit line set includes a second bit line and a second complementary bit line. At least one of the second bit line and the second complementary bit line is disposed on the backside of the substrate.

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Background/Summary

BACKGROUND

[0001] The semiconductor integrated circuit industry has experienced rapid growth in recent years. Technological advances in semiconductor materials and design have produced increasingly compact and complex circuits. Such material and design advances have been made possible as technologies related to processing and manufacturing have also undergone technical advances. As the semiconductor industry has progressed into advanced technology process nodes in pursuit of smaller product scales and multiple functions, various approaches have been studied and obstacles of electrical resistance and coupling effect of signals of a memory device have been encountered.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It should be noted that, in accordance with standard practice in the industry, various features are not drawn to scale. In fact, dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0003] FIG. 1 is a schematic 3D diagram of a semiconductor structure in accordance with some embodiments of the present disclosure.

[0004] FIG. 2 is a schematic cross-sectional diagram of the semiconductor structure of FIG. 1 along a line A-A' in accordance with some embodiments of the present disclosure.

[0005] FIG. 3 is a schematic 3D diagram of a semiconductor structure in accordance with some embodiments of the present disclosure.

[0006] FIG. 4 is a schematic cross-sectional diagram of the semiconductor structure of FIG. 3 along a line B-B' in accordance with some embodiments of the present disclosure.

[0007] FIGS. 5 to 13 are schematic 3D diagrams of semiconductor structures for illustrations of different arrangements of signal lines in order to achieve a data storage device in accordance with different embodiments of the present disclosure.

[0008] FIG. 14 is a schematic 3D diagram of a semiconductor structure in accordance with some embodiments of the present disclosure.

[0009] FIG. 15 is a schematic top-view perspective of the semiconductor structure shown in FIG. 14 in accordance with some embodiments of the present disclosure.

[0010] FIG. 16 is a schematic bottom-view perspective of the semiconductor structure shown in FIG. 14 in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0011] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of elements and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0012] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “over,”

“upper,” “on” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0013] As used herein, although the terms such as “first,” “second” and “third” describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another. The terms such as “first,” “second” and “third” when used herein do not imply a sequence or order unless clearly indicated by the context. In addition, the term “source/drain region” or “source/drain regions” may refer to a source or a drain, individually or collectively dependent upon the context.

[0014] Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the disclosure are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from normal deviation found in the respective testing measurements. Also, as used herein, the terms “substantially,” “approximately” and “about” generally mean within a value or range that can be contemplated by people having ordinary skill in the art. Alternatively, the terms “substantially,” “approximately” and “about” mean within an acceptable standard error of the mean when considered by one of ordinary skill in the art. People having ordinary skill in the art can understand that the acceptable standard error may vary according to different technologies. Other than in the operating/working examples, or unless otherwise expressly specified, all of the numerical ranges, amounts, values and percentages, such as those for quantities of materials, durations of time, temperatures, operating conditions, ratios of amounts, and the likes thereof disclosed herein, should be understood as modified in all instances by the terms “substantially,” “approximately” or “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the present disclosure and attached claims are approximations that can vary as desired. At the very least, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Ranges can be expressed herein as from one endpoint to another endpoint or between two endpoints. All ranges disclosed herein are inclusive of the endpoints, unless specified otherwise.

[0015] FIG. 1 is a schematic three-dimensional (3D) diagram of a semiconductor structure **601**, and FIG. 2 is a schematic cross-sectional diagram of the semiconductor structure **601** of FIG. 1 along a line A-A' in accordance with some embodiments of the present disclosure. The semiconductor structure **601** includes a substrate layer **101** and an active layer **105** disposed over the substrate layer **101**. The substrate layer **101** and the active layer **105** may collectively be referred to as a substrate **10**. The active layer **105** is proximal to a front side **S1** of the substrate **10**, and the substrate layer **101** is proximal to a backside **S2** of the substrate **10**. In some embodiments, a bottom surface of the substrate layer **101** corresponds to a back surface of the substrate **10**, wherein the back surface faces the backside **S2** of the substrate **10**.

[0016] In some embodiments, the active layer **105** is a layer of active regions of the substrate **10**. In some embodiments, a plurality of fin structures, nanosheets or nanowires may be formed in the active layer **105**. The active layer **105** can include multiple portions (e.g., **105a**, **105b**, **105c** and **105d**), and each portion can be an active region. In some embodiments, the active regions **105a** and **105b** are substantially aligned and extend along a same direction. In some embodiments, the active regions **105c** and **105d** are substantially aligned and extend along a same direction. In some embodiments, the active region **105a** is adjacent to and substantially parallel to the active region **105c**. In some embodiments, the active region **105b** is adjacent to and substantially parallel to the active region **105d**. In some embodiments, each of the active regions **105a**, **105b**, **105c** and **105d**

includes at least one transistor (e.g., T1, T2, T3 or T4). The substrate **10** can include a bulk semiconductor material, such as silicon, or another semiconductor material, such as silicon germanium, silicon carbide, gallium arsenide, or the like. The substrate **10** may be of a first conductivity type, e.g., a P-type semiconductive substrate (acceptor type), or a second conductivity type, e.g., an N-type semiconductive substrate (donor type).

[0017] The transistors T1, T2, T3 and T4 can include one or more types of transistors, such as a multi-gate transistor, a gate-all-around field-effect transistor (GAAFET), a fin field-effect transistor (FinFET), a vertical transistor, a nanosheet transistor, a nanowire transistor, or a combination thereof. Each of the transistors T1, T2, T3 and T4 includes a pair of source/drain (S/D) structures (e.g., **103** and **104**) surrounding the active layer **105**, and a gate structure disposed between the pair of S/D structures and surrounding the active layer **105** (e.g., fin structures, nanosheets or nanowires of the active layer **105** between the S/D structures). In some embodiments, each of the S/D structures includes an upper portion disposed above the active layer **105** and a lower portion disposed below the active layer **105**. In some embodiments, the S/D structures are epitaxial structures and include one or more semiconductor materials, such as polysilicon, silicon germanium, silicon carbide, gallium arsenide, or the like.

[0018] In some embodiments, the transistor T1 is disposed in the active region **105a** and includes S/D structures **103a** and **103b** and a gate structure **111a** disposed between the S/D structures **103a** and **103b**. In some embodiments, the transistor T2 is disposed in the active region **105b** and includes S/D structures **103c** and **103d** and a gate structure **111b** disposed between the S/D structures **103c** and **103d**. In some embodiments, the transistor T3 is disposed in the active region **105c** and includes S/D structures **104a** and **104b** and a gate structure **111c** disposed between the S/D structures **104a** and **104b**. In some embodiments, the transistor T4 is disposed in the active region **105d** and includes S/D structures **104c** and **104d** and a gate structure **111d** disposed between the S/D structures **104c** and **104d**. The S/D structures **103a** and **103b** and the gate structure **111a** may be arranged along a first horizontal direction (e.g., X direction).

[0019] The S/D structures and the gate structure of each of the transistors T2, T3 and T4 may be arranged along the first horizontal direction. In some embodiments, the transistors T1 and T2 are arranged along the first horizontal direction. In some embodiments, the transistors T1 and T3 are arranged along a second horizontal direction (e.g., Y direction), which is substantially perpendicular to the first horizontal direction. In some embodiments, the S/D structures **103a** and **104a** are aligned along the second horizontal direction. In some embodiments, the S/D structures **103b** and **104b** are aligned along the second horizontal direction. In some embodiments, the S/D structures **103c** and **104c** are aligned along the second horizontal direction. In some embodiments, the S/D structures **103d** and **104d** are aligned along the second horizontal direction.

[0020] The gate structures **111a** and **111c** can be physically connected. In some embodiments, the gate structures **111a** and **111c** are a monolithic structure. The gate structures **111b** and **111d** can be physically connected. In some embodiments, the gate structures **111b** and **111d** are a monolithic structure. It should be noted that one of the pair of the S/D structures **103a** and **103b** can be a source structure, and another of the pair of the S/D structures **103a** and **103b** can be a drain structure. Similarly, the S/D structure **103c**, **104a**, or **104c** can be a source structure or a drain structure, and the S/D structure **103d**, **104b**, or **104d** can be a drain structure or a source structure.

[0021] Referring to FIG. 2, the semiconductor structure **601** may further include a front-side redistribution structure **20** disposed on the front side S1 of the substrate **10** and a backside redistribution structure **30** disposed on the backside S2 of the substrate **10**. In other words, the front-side redistribution structure **20** is disposed on a side of the substrate **10** opposite to the backside redistribution structure **30**.

[0022] Each of the front-side redistribution structure **20** and the backside redistribution structure **30** may include multiple metal line layers and multiple metal via layers. For example, the front-side redistribution structure **20** includes metal line layers M0 to Mn, wherein the metal line layer M0 is

a first metal line layer disposed over and closest to the substrate **10** of the front-side redistribution structure **20**, and the metal line layer Mn is a top-most metal line layer disposed over and farthest from the substrate **10** of the front-side redistribution structure **20**. For another example, the backside redistribution structure **30** includes metal line layers BM0 to BMn, wherein the metal line layer BM0 is a first metal line layer disposed below and closest to the substrate **10** of the backside redistribution structure **30**, and the metal line layer BMn is a bottom-most metal line layer disposed below and farthest from the substrate **10** of the front-side redistribution structure **20**. The multiple metal via layers are arranged alternately between the metal line layers for electrical connection between the metal line layers. In some embodiments, each metal line layer is formed of metal lines and an intermetal dielectric (IMD) layer surrounding the metal lines. In some embodiments, each metal via layer is formed of metal vias and an IMD layer surrounding the metal vias.

[0023] It should be noted that the S/D structures **103** and **104** can be physically separated or electrically isolated from one another by isolations made of one or more dielectric materials. Similarly, the gate structures **111a** and **111c**, and the gate structures **111b** and **111d**, can be physically separated or electrically isolated from each other by the isolations. The isolations or the dielectric materials are not shown in the figures for a purpose of simplicity and brevity.

[0024] As shown in FIG. 2, in some embodiments, a metal line **221** disposed in the metal line layer M0 electrically connects to the gate structure **111a** through a via **211** disposed in a via layer between the metal line layer M0 and the gate structure **111a**. In some embodiments, the metal line **221** in the metal line layer M0 electrically connects to the S/D structure **103d** through a via **212** disposed in the same via layer as the via **211**. In some embodiments, a metal line **321** disposed in the metal line layer BM0 electrically connects to the gate structure **111a** through a via **311** disposed in a via layer between the metal line layer BM0 and the gate structure **111a**. In some embodiments, the via **311** penetrates the substrate layer **101** to electrically connect to the gate structure **111a** from the backside S2 of the substrate **10**. In some embodiments, the metal line **321** in the metal line layer BM0 electrically connects to the S/D structure **103a** through a via **312** disposed in the same via layer as the via **311**. In some embodiments, the via **312** penetrates the substrate layer **101** to electrically connect to the S/D structure **103a** from the backside S2 of the substrate **10**. It should be noted that routings and electrical connections between the different transistors T1 and T2 can be adjusted through the front-side redistribution structure **20** and the backside redistribution structure **30** according to different applications, and that FIG. 2 is for a purpose of illustration but is not intended to limit the present disclosure. In addition, a number of transistors in one active region of the active layer **105** can be adjusted according to different applications. The transistors (e.g., T1, T2, T3 and T4) collectively correspond to a data storage element, for example, a static random-access memory (SRAM) device, and signals for read and write (e.g., bit line signal, complementary bit line signal, and word line signal) can be transmitted through the front-side redistribution structure **20** and the backside redistribution structure **30**.

[0025] It should be noted that a height of the gate structure **111a** and a height of the gate structure **111b** along vertical direction (e.g., Z direction) can be substantially equal or different. In some embodiments, the heights of the gate structures **111a** and **111b** are different. As shown in FIG. 2, the height of the gate structure **111a** is less than the height of the gate structure **111b** along the vertical direction. In some embodiments, the gate structure **111a** is above and separated from the substrate layer **101**. In some embodiments, the gate structure **111b** contacts the substrate layer **101**. However, the present disclosure is not limited thereto. In other embodiments, the heights of the gate structure **111a** and the height of the gate structure **111b** are substantially equal for a purpose of ease of manufacturing process.

[0026] FIG. 3 is a schematic 3D diagram of a semiconductor structure **602**, and FIG. 4 is a schematic cross-sectional diagram of the semiconductor structure **602** of FIG. 3 along a line B-B' in accordance with some embodiments of the present disclosure. The semiconductor structure **602** can be similar to the semiconductor structure **601**, except that the semiconductor structure **602** includes

another active layer **106** disposed above the active layer **105**, and the transistors **T3** and **T4** are formed in and surround the active layer **106**. For a purpose of brevity, in the following illustration, only differences between the semiconductor structures **601** and **602** are described. It should be noted that such omission of similar features of the semiconductor structures **601** and **602** is not intended to limit the present disclosure.

[0027] The substrate layer **101** and the active layers **105** and **106** are collectively referred to as the substrate **10**. The active layer **105** is disposed between the active layer **106** and the substrate layer **101** along a vertical direction (e.g., Z direction). The active layer **106** is proximal to the front side **S1** of the substrate **10**, and the substrate layer **101** is proximal to the backside **S2** of the substrate **10**. In other words, the different active layers **105** and **106** are at different elevations above the substrate layer **101**, and the active regions in a same active layer are disposed at a same elevation.

[0028] Similar to the active layer **105**, the active layer **106** can include multiple active regions. In some embodiments, the active layer **105** includes active regions **105a** and **105b**, and the active layer **106** includes active regions **106a** and **106b** substantially aligned with the active regions **105a** and **105b**, respectively, along the vertical direction. In some embodiments, the active regions **105a** and **105b** are substantially parallel. In some embodiments, the active regions **105a** and **105b** are arranged along the second horizontal direction, and each of the active regions **105a** and **105b** extends along the first horizontal direction, which is substantially perpendicular to the second horizontal direction and the vertical direction.

[0029] The transistors **T1**, **T2**, **T3** and **T4** can include one or more types of transistors, such as a multi-gate transistor, a gate-all-around field-effect transistor (GAAFET), a fin field-effect transistor (FinFET), a vertical transistor, a nanosheet transistor, a nanowire transistor, or a combination thereof. In some embodiments, the transistors **T1**, **T2**, **T3** and **T4** are complementary field-effect transistors (CFETs).

[0030] In some embodiments, the gate structures of the transistors **T1**, **T2**, **T3** and **T4** are all connected. In some embodiments, the gate structures of the transistors **T1**, **T2**, **T3** and **T4** are a monolithic structure. In some embodiments, the transistors **T1**, **T2**, **T3** and **T4** have a common gate structure **111**. In some embodiments, the common gate structure **111** extends across the active layers **105** and **106**. In some embodiments, the common gate structure **111** surrounds the active layers **105** and **106** respectively. In some embodiments, the common gate structure **111** extends from above the active layer **106** to below the substrate layer **101**.

[0031] In some embodiments, the transistor **T1** is disposed in the active region **105a**, and the S/D structures **103a** and **103b** of the transistor **T1** are disposed in and surround the active region **105a** on two opposite sides of the common gate structure **111**. In some embodiments, the transistor **T2** is disposed in the active region **105b**, and the S/D structures **103c** and **103d** of the transistor **T2** are disposed in and surround the active region **105b** on two opposite sides of the common gate structure **111**. In some embodiments, the transistor **T3** is disposed in the active region **106a**, and the S/D structures **104a** and **104b** of the transistor **T3** are disposed in and surround the active region **106a** on two opposite sides of the common gate structure **111**. In some embodiments, the transistor **T4** is disposed in the active region **106b**, and the S/D structures **104c** and **104d** of the transistor **T4** are disposed in and surround the active region **106b** on two opposite sides of the common gate structure **111**. In some embodiments, the S/D structures **103a** and **104a** are vertically aligned. In some embodiments, the S/D structures **103b** and **104b** are vertically aligned. In some embodiments, the S/D structures **103c** and **104c** are vertically aligned. In some embodiments, the S/D structures **103d** (not visible in FIG. 3) and **104d** are vertically aligned. The S/D structures **103a** to **103d** and **104a** to **104d** may be electrically isolated from one another by an isolation (or dielectric material) or electrically connected through an inter-S/D structure **107** as shown in FIG. 4. The inter-S/D structure **107** can include conductive material or semiconductor material, such as polysilicon, germanium, silicon germanium, or other suitable materials.

[0032] As shown in FIG. 4, similar to the embodiments shown in FIG. 2, a metal line **221** is

disposed in a metal line layer **M0**, and a metal line **321** is disposed in a metal line layer **BM0** for electrical connection and circuit routing to achieve a data storage element. In some embodiments, the common gate structure **111** is electrically connected to the metal line **221** from the front side **S1** of the substrate **10** through a via **211**. In some embodiments, the common gate structure **111** is electrically connected to the metal line **321** from the backside **S2** of the substrate **10** through a via **311**. In some embodiments, each of the S/D structures **103a** to **103d** includes an upper portion disposed above the active layer **105**. In some embodiments, each of the S/D structures **103a** to **103d** includes a lower portion disposed below the active layer **105**. In some embodiments, each of the S/D structures **104a** to **104d** includes an upper portion disposed above the active layer **106**. In some embodiments, each of the S/D structures **104a** to **104d** includes a lower portion disposed below the active layer **106**. In some embodiments, the inter-S/D structure **107** is entirely disposed between the active layers **105** and **106**. In some embodiments, the inter-S/D structure **107** is entirely disposed between the S/D structures **104a** and **103a**.

[0033] FIGS. 5 to 14 are schematic 3D diagrams of semiconductor structures **603** to **612** for illustrations of different arrangements of signal lines in order to achieve a data storage device with improved resistance and/or reduced coupling effect in accordance with different embodiments of the present disclosure. Each of the semiconductor structures **603** to **612** includes the semiconductor structure **602** shown in FIG. 3 as exemplary illustration. In other embodiments, the semiconductor structure **601** can be applied to achieve a data storage device, and the configuration is not limited herein.

[0034] As shown in FIG. 5, in some embodiments, metal lines **222** and **223** are disposed in the metal line layer **M0** above the transistors **T1** to **T4** (collectively referred to as a transistor array **40** for a purpose of illustration), and metal lines **322** and **323** are disposed in the metal line layer **BM0** below the substrate layer **101**. The metal lines **222**, **223**, **322** and **323** are for transmission of bit line signals and complementary bit line signals. In some embodiments, the metal lines **222**, **223**, **322** and **323** are substantially parallel to one another. In some embodiments, each of the metal lines **222**, **223**, **322** and **323** extends along the first horizontal direction. In some embodiments, the metal line **222** is substantially symmetrical to the metal line **322** with respect to the transistor array **40**. In some embodiments, the metal line **223** is substantially symmetrical to the metal line **323** with respect to the transistor array **40**.

[0035] In some embodiments, different bit line signals may be designed to be transmitted on two opposite sides of the substrate **10** (or two opposite sides of the transistor array **40**) for a purpose of reduction of electrical resistance. In some embodiments, the metal lines **222** and **223** are for transmission of a bit line signal and its complementary signal. In some embodiments, the metal line **222** is a bit line, and the metal line **223** is a complementary bit line (or a bit line bar) of the metal line **222**. In some embodiments, the metal lines **322** and **323** are for transmission of another bit line signal and its complementary signal. In some embodiments, the metal line **322** is a bit line, and the metal line **323** is a complementary bit line (or a bit line bar) of the metal line **322**.

[0036] Since the front-side redistribution structure **20** (including the metal line layer **M0**) is formed separately from the backside redistribution structure **30** (including the metal line layer **BM0**), it is easy to integrate different conductive materials of the metal lines in the front-side redistribution structure **20** and the backside redistribution structure **30**. In addition, formation of the front-side redistribution structure **20** should be precisely controlled in order to prevent damage of electrical elements formed by the front-end of line (FEOL) manufacturing process. It is difficult to integrate new and/or different materials of metal lines of the front-side redistribution structure **20**. However, it is easier to introduce new or different materials of metal lines of the backside redistribution structure **30**. Therefore, an electrical resistance of a bit line and a complementary bit line can be adjusted or reduced by applying a material different from that of the front-side redistribution structure **20**. In some embodiments, a material of the metal lines **322** and **323** is different from a material of the metal lines **222** and **223**. In some embodiments, the material of the metal lines **322**

and **323** has an electrical resistance substantially lower than a resistance of the material of the metal lines **222** and **223**.

[0037] As shown in FIG. 5, in some embodiments, a bit line signal and its complementary bit line signal are arranged to be transmitted on two opposite sides of the substrate **10** (or two opposite sides of the transistor array **40**) for a purpose of reduction of a cross-coupling effect. In some embodiments, the metal lines **222** and **322** are for transmission of a bit line signal and its complementary signal. In some embodiments, the metal lines **222** and **322** are vertically aligned. In some embodiments, the metal line **222** is a bit line, and the metal line **322** is a complementary bit line (or a bit line bar) of the metal line **222**. In some embodiments, the metal lines **223** and **323** are for transmission of another bit line signal and its complementary signal. In some embodiments, the metal lines **223** and **323** are vertically aligned. In some embodiments, the metal line **223** is a bit line, and the metal line **323** is a complementary bit line (or a bit line bar) of the metal line **223**. In addition, a material of the metal lines **322** and **323** can be different from a material of the metal lines **222** and **223** as illustrated above, and thus, an overall electrical resistance of the bit lines and the complementary bit lines can be also reduced.

[0038] As shown in FIG. 5, in some embodiments, a bit line pair (including a bit line and a complementary bit line) and a fly bit line pair (including a fly bit line and a complementary fly bit line) are arranged to be disposed on two opposite sides of the substrate **10** (or two opposite sides of the transistor array **40**) for a purpose of improvement of routing resource in the front-side redistribution structure **20**. The fly bit line pair is for a purpose of reduction of electrical resistance of the bit line pair. In some embodiments, the fly bit line is for transmission of a signal identical to a bit line signal of the bit line. In some embodiment, the fly bit line is for transmission of a signal identical to a complementary signal of a complimentary bit line. In some embodiments, the bit line pair includes two adjacent bit lines (e.g., **222** and **223** or **322** and **323**). In some embodiments, the fly bit line pair includes two adjacent bit lines (e.g., **322** and **323** or **222** and **223**), which are disposed opposite to the bit line pair with respect to substrate **10** (or two opposite sides of the transistor array **40**) In a comparative embodiment when the fly bit line pair and the bit line pair are both disposed in the front-side redistribution structure **20**, the fly bit line pair is disposed in two metal line layers above the bit line pair (bit lines and word lines are alternately arranged in different metal line layers of the front-side redistribution structure **20**, and therefore the fly bit line pair cannot be arranged in a metal line layer directly over the metal line layer of the bit line pair). Therefore, a routing design of the comparative embodiment is more complicated than that of the embodiments shown in FIG. 5. For a purpose of increase of routing resource in the front-side redistribution structure **20**, the fly bit lines can be disposed in the backside redistribution structure **30**, and therefore routing resources in the front-side redistribution structure **20** can be improved.

[0039] In some embodiments, the metal lines **222** and **223** are a bit line pair. In some embodiments, the metal lines **322** and **323** are a fly bit line pair. In some embodiments, the metal lines **322** and **323** are a bit line pair. In some embodiments, the metal lines **222** and **223** are a fly bit line pair.

[0040] FIGS. **6** to **8** are schematic 3D diagrams of semiconductor structures **604** to **606** including a pair of bit lines, a pair of fly bit lines, and a pair of complementary bit lines in accordance with different embodiments of the present disclosure.

[0041] In some embodiments as shown in FIG. **6**, the semiconductor structure **604** is similar to the semiconductor structure **603** with additional metal lines **225** and **226** disposed in the metal line layer **M0**. The metal lines **225** and **226** can for transmission of signals other than bit line signals for a purpose of prevention of affecting signal judgement to the bit line signals and the complementary bit line signals through the metal lines **222** and **223**. In some embodiments, the metal lines **226** and **223** are a first bit line pair, the metal lines **222** and **225** are a second bit line pair, and the metal lines **322** and **323** are a fly bit line pair, which is paired to the first bit line pair or the second bit line pair. In some embodiments, only one of the first bit line pair and the second bit line pair has a paired fly bit line pair disposed in the backside redistribution structure **30**. It should be noted that the paired

fly bit line pair may or may not be vertically aligned with it paired bit line pair. In some embodiments, the metal lines **322** and **323** are not vertically aligned with the paired metal lines **226** and **223** or **222** and **225** as shown in FIG. **6**. In some embodiments, the metal lines **322** and **323** are vertically overlapped by the paired metal lines **226** and **223** respectively. In some embodiments, the metal lines **322** and **323** are vertically overlapped by the paired metal lines **222** and **225** respectively.

[0042] In some embodiments as shown in FIG. **7**, the semiconductor structure **605** is similar to the semiconductor structure **603**, except that the semiconductor structure **605** includes additional metal lines **225** and **226** disposed in a metal line layer **M2** above the metal line layer **M0**. The metal lines **225** and **226** can be bit lines, fly bit lines or complementary bit lines according to different arrangements of the metal lines **222**, **223**, **322** and **323** as depicted in FIG. **5** and described in relevant paragraphs above. As mentioned above, repeated description is omitted herein for a purpose of brevity.

[0043] In some embodiments as shown in FIG. **6**, the semiconductor structure **604** is similar to the semiconductor structure **603** with additional metal lines **225** and **226** disposed in the metal line layer **M0**. The metal lines **225** and **226** can for transmission of signals other than bit line signals for a purpose of prevention of affecting signal judgement to the bit line signals and the complementary bit line signals through the metal lines **222** and **223**. In some embodiments, the metal lines **226** and **223** are a first bit line pair, the metal lines **222** and **225** are a second bit line pair, and the metal lines **322** and **323** are a fly bit line pair, which is paired to the first bit line pair or the second bit line pair. In some embodiments, only one of the first bit line pair and the second bit line pair has a paired fly bit line pair disposed in the backside redistribution structure **30**. It should be noted that the paired fly bit line pair may or may not be vertically aligned with it paired bit line pair. In some embodiments, the metal lines **322** and **323** are not vertically aligned with the paired metal lines **226** and **223** or **222** and **225** as shown in FIG. **6**. In some embodiments, the metal lines **322** and **323** are vertically overlapped by the paired metal lines **226** and **223** respectively. In some embodiments, the metal lines **322** and **323** are vertically overlapped by the paired metal lines **222** and **225** respectively.

[0044] FIGS. **9** to **11** are schematic 3D diagrams of semiconductor structures **607** to **609** for illustration of different arrangements of word lines in order to achieve a data storage device with improved resistance and/or reduced coupling effect in accordance with different embodiments of the present disclosure.

[0045] As shown in FIG. **9**, in some embodiments, a metal line **331** is disposed in a metal line layer **BM1** below the substrate layer **101** for a purpose of transmission of word line signals through the backside redistribution structure **30**. In some embodiments, the metal line layers for transmission of word line signals are referred to as word line layers, and the metal line layers for transmission of bit line signals and complementary bit line signals are referred to as bit line layers. The word line layers and the bit line layers are alternately arranged along the vertical direction in the front-side redistribution structure **20** or the backside redistribution structure **30** starting with a bit line layer (i.e., a first bit line layer is the metal line layer **M0** or **BM0**, and a first word line layer is the metal line layer **M1** or **BM0**). In some embodiments, the metal line **331** extends along the second horizontal direction. In some embodiments, the metal line **331** is substantially aligned with the common gate structure **111**. In some embodiments, the metal line **331** is substantially parallel to the common gate structure **111**. The arrangement of the metal line **331** (functioning as a word line) in the metal line layer **BM1** can provide increase of routing resources in the front-side redistribution structure **20**. In addition, an electrical resistance of the metal line **331** can be reduced compared to a word line disposed in the front-side redistribution structure **20**.

[0046] As shown in FIG. **10**, the semiconductor structure **608** can be similar to the semiconductor **607**, except that the semiconductor structure **608** includes an additional metal line **231** disposed in a metal line layer **M1** above the metal line layer **M0** shown in FIG. **5**. In some embodiments, the

metal lines **231** and **331** are for a purpose of transmission of different word line signals through the front-side redistribution structure **20** and the backside redistribution structure **30**. In some embodiments, the metal lines **231** and **331** referred to as double word lines, and the metal lines **231** and **331** are configured to transmit identical signals for a purpose of reduction of an overall electrical resistance of word lines. In some embodiments, the metal lines **231** and **331** are in parallel connection when they are double word lines. In some embodiments, the metal line **231** extends along the second horizontal direction. In some embodiments, the metal line **231** is substantially aligned with the common gate structure **111**. In some embodiments, the metal line **231** is substantially parallel to the common gate structure **111**. In some embodiments, the metal line **231** is substantially symmetrical to the metal line **331** with respect to the common gate structure **111**.

[0047] As shown in FIG. **11**, the semiconductor structure **609** can be similar to the semiconductor structure **608**, except that the semiconductor structure **609** includes an additional metal line **251** disposed in a metal line layer M3 above the metal line layer M1. Similar to the purposes described above, in some embodiments, the metal lines **231**, **331** and **251** are for a purpose of transmission of different word line signals through the front-side redistribution structure **20** and the backside redistribution structure **30**. In some embodiments, the metal lines **231**, **331** and **251** are for a purpose of reduction of an overall electrical resistance of word lines. Since the metal lines **231**, **331** and **251** are disposed in different front-side and backside redistribution structure **30**s on opposite sides of the substrate **10**, a complexity of routing can be reduced, and the routing resources can be increased. In addition, an overall electrical resistance of word lines can be reduced compared to that of a configuration in which all of the word lines are disposed on one side of the substrate **10**.

[0048] FIG. **12** is a schematic 3D diagram of a semiconductor structure **610** showing positional relationships between bit lines and word lines in accordance with some embodiments of the present disclosure. In some embodiments, the semiconductor structure **610** includes the metal lines **222**, **223**, **322** and **323** shown in FIG. **5** and the metal line **331** shown in FIG. **9** for a purpose of illustration. In some embodiments, the metal line **331** is substantially perpendicular to one of the metal lines **222**, **223**, **322** or **323**. In some embodiments, the metal line **331** is electrically isolated or physically separated from the metal lines **322** and **323**, for example, by an IMD layer (not shown) of the backside redistribution structure **30**.

[0049] FIG. **13** is a schematic 3D diagram of a semiconductor structure **611** showing positional relationships between bit lines and word lines in accordance with some embodiments of the present disclosure. In some embodiments, the semiconductor structure **611** includes the metal lines **222**, **223**, **322** and **323** shown in FIG. **5**, the metal line **331** shown in FIG. **9**, and the metal line **231** shown in FIG. **10** for a purpose of illustration. In some embodiments, the metal lines **231** and **331** are substantially perpendicular to one of the metal lines **222**, **223**, **322** and **323**. In some embodiments, the metal line **331** is electrically isolated or physically separated from the metal lines **322** and **323**, for example, by an IMD layer (not shown) of the backside redistribution structure **30**. Similarly, in some embodiments, the metal line **231** is electrically isolated or physically separated from the metal lines **222** and **223**. The metal lines shown in FIGS. **5** to **8** can be applied to any of the structures shown in FIGS. **9** to **11**, and description is omitted herein for a purpose of brevity. However, such omission is not intended to limit the present disclosure.

[0050] FIG. **14** is a schematic 3D diagram of a semiconductor structure **612**, FIG. **15** is a schematic top-view perspective of the semiconductor structure **612** shown in FIG. **14**, and FIG. **16** is a schematic bottom-view perspective of the semiconductor structure **612** shown in FIG. **14** in accordance with some embodiments of the present disclosure.

[0051] As shown in FIG. **14**, the semiconductor structure **612** can be similar to the semiconductor structure **603**, except the semiconductor structure **612** can include a different number of transistors in the transistor array **40**, and additional metal lines **224** and **324** for power/ground connection. In some embodiments, the transistor array **40** includes S/D structures **103a**, **103b** and **103c** disposed in and surrounding an active layer **105**. In some embodiments, the transistor array **40** includes gate

structures **111a** and **111b** disposed in and surrounding the active layer **105**. In some embodiments, the gate structure **111a** is between the S/D structures **103a** and **103b** along a first horizontal direction (e.g., X direction). In some embodiments, the gate structure **111b** is between the S/D structures **103b** and **103c** along the first horizontal direction.

[0052] In some embodiments, the transistor array **40** includes S/D structures **104a**, **104b** and **104c** disposed in and surrounding an active layer **106**, which is disposed above the active layer **105**. In some embodiments, the S/D structures **104a**, **104b** and **104c** are disposed above and vertically aligned with the S/D structures **103a**, **103b** and **103c**, respectively. In some embodiments, the transistor array **40** includes gate structures **111c** and **111d** disposed in and surrounding the active layer **106**. In some embodiments, the gate structures **111c** and **111d** are disposed above and vertically aligned with the gate structures **111a** and **111b**. In some embodiments, the gate structure **111c** is electrically isolated or physically separated from the gate structure **111a** by an isolation **112a**. In some embodiments, the gate structure **111d** is electrically isolated or physically separated from the gate structure **111b** by an isolation **112b**. In some embodiments, the gate structure **111c** is between the S/D structures **104a** and **104b** along the first horizontal direction. In some embodiments, the gate structure **111d** is between the S/D structures **104b** and **104c** along the first horizontal direction.

[0053] In some embodiments, the transistor array **40** includes inter-S/D structures **107a**, **107b** and **107c** for electrical connections between the S/D structures **103a** and **104a**, **103b** and **104b**, and **103c** and **104c**, respectively. In some embodiments, the inter-S/D structures **107a**, **107b** and **107c** are disposed between the active layers **105** and **106** along the vertical direction. In some embodiments, the inter-S/D structure **107a** is within a vertical projection of the S/D structure **103a** or **104a**. In some embodiments, the inter-S/D structure **107b** is within a vertical projection of the S/D structure **103b** or **104b**. In some embodiments, the inter-S/D structure **107c** is within a vertical projection of the S/D structure **103c** or **104c**.

[0054] The metal lines **222**, **223**, **322** and **323** are bit lines, complementary bit lines, and/or fly bit lines as described above, and repeated description is omitted herein. In some embodiments, the metal line **222** is electrically connected to the S/D structure **104a** from the front side **S1** of the substrate **10** through a via **212a** disposed between the transistor array **40** and the metal line layer **M0**. In some embodiments, the metal line **223** is electrically connected to the S/D structure **104c** from the front side **S1** of the substrate **10** through a via **212c** disposed between the transistor array **40** and the metal line layer **M0**. In some embodiments, the metal line **322** is electrically connected to the S/D structure **103a** from the backside **S2** of the substrate **10** through a via **312a** disposed between the transistor array **40** and the metal line layer **BM0**. In some embodiments, the metal line **323** is electrically connected to the S/D structure **103c** from the backside **S2** of the substrate **10** through a via **312c** disposed between the transistor array **40** and the metal line layer **BM0**.

[0055] The metal line **224** can be disposed between the metal lines **222** and **223** in the metal line layer **M0**. In some embodiments, the metal line **224** is for power/ground connection to the transistor array **40**. In other words, a power/ground line (i.e. **224**) is disposed between, for example, adjacent bit lines (e.g., **222** and **223**) in the metal line layer **M0** along a second horizontal direction (e.g., Y direction). In some embodiments, the metal line **224** is electrically connected to the S/D structure **104b** from the front side **S1** of the substrate **10** through a via **212b** disposed between the transistor array **40** and the metal line layer **M0**. The metal line **324** can be disposed between the metal lines **322** and **323** in the metal line layer **BM0**. In some embodiments, the metal line **324** is for power/ground connection to the transistor array **40**. In other words, a power/ground line (i.e. **224**) is disposed between, for example, adjacent complementary bit lines (e.g., **322** and **323**) in the metal line layer **BM0** along the second horizontal direction. In some embodiments, the metal line **324** is electrically connected to the S/D structure **103b** from the backside **S2** of the substrate **10** through a via **312b** disposed between the transistor array **40** and the metal line layer **BM0**.

[0056] For ease of formation, widths of metal lines in a same metal line layer are substantially

equal. In some embodiments, as shown in FIG. 15, the metal lines 222, 223, and 224 have a substantially equal width W20, wherein the width W20 is measured along the second horizontal direction. In some embodiments, as shown in FIG. 16, the metal lines 322, 323, and 324 have a substantially equal width W30, wherein the width W30 is measured along the second horizontal direction. Since routings in the front-side redistribution structure 20 can be much more complex than routings in the backside redistribution structure 30, a width of a metal line in a metal line layer (e.g., M0) in the front-side redistribution structure 20 is substantially less than a width of a metal line in a corresponding metal line layer (e.g., BM0) in the backside redistribution structure 30. [0057] Therefore, the present disclosure provides a semiconductor structure having a data storage element (e.g., a static random-access memory (SRAM)) with memory signal transmission through a backside redistribution structure. As a result, a cross-coupling effect can be reduced, and routing resources can be increased. In addition, an overall electrical resistance may also be reduced since it is easier to change materials of metal lines in the backside redistribution structure than to change materials of metal lines in the front-side redistribution structure.

[0058] In accordance with some embodiments of the disclosure, a semiconductor structure is provided. The semiconductor structure includes a substrate having a front side and a backside opposite to the front side, wherein the substrate includes a substrate layer, a first active layer disposed above the substrate layer, and a second active layer disposed above the first active layer, the second active layer is proximal to the front side, and the substrate layer is proximal to the backside; a data storage element, disposed in and surrounding the first active layer and the second active layer; a first bit line set, including a first bit line and a first complementary bit line, wherein at least one of the first bit line and the first complementary bit line is disposed on the front side of the substrate above the second active layer; and a second bit line set, including a second bit line and a second complementary bit line, wherein at least one of the second bit line and the second complementary bit line is disposed on the backside of the substrate below the substrate layer.

[0059] In accordance with some embodiments of the disclosure, a semiconductor structure is provided. The semiconductor structure includes a substrate having a front side and a backside opposite to the front side, wherein the substrate includes a substrate layer, a first active layer disposed above the substrate layer, and a second active layer disposed above the first active layer, the second active layer is proximal to the front side, and the substrate layer is proximal to the backside; a data storage element, disposed in and surrounding the first active layer and the second active layer; and a first word line, disposed on the backside of the substrate below the substrate layer.

[0060] In accordance with some embodiments of the disclosure, a semiconductor structure is provided. The semiconductor structure includes a substrate having a front side and a backside opposite to the front side, wherein the substrate includes a substrate layer proximal to the backside; a plurality of complementary FETs (CFETs), disposed on the substrate layer, wherein the CFETs are proximal to the front side of the substrate and collectively correspond to a data storage element; a first bit line set, including a first bit line and a first complementary bit line, wherein at least one of the first bit line and the first complementary bit line is disposed on the front side of the substrate above the CFETs; a second bit line set, including a second bit line and a second complementary bit line, wherein at least one of the second bit line and the second complementary bit line is disposed on the backside of the substrate below the substrate layer; and a first power/ground line, disposed on the backside of the substrate below the substrate layer at an elevation same as that of the at least one of the second bit line and the second complementary bit line.

[0061] The foregoing outlines features of several embodiments so that those skilled in the art may better understand aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent

constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor structure, comprising: a substrate, having a front side and a backside opposite to the front side, wherein the substrate includes a substrate layer, a first active layer disposed above the substrate layer, and a second active layer disposed above the first active layer, the second active layer is proximal to the front side, and the substrate layer is proximal to the backside; a data storage element, disposed in and surrounding the first active layer and the second active layer; a first bit line set, including a first bit line and a first complementary bit line, wherein at least one of the first bit line and the first complementary bit line is disposed on the front side of the substrate above the second active layer; and a second bit line set, including a second bit line and a second complementary bit line, wherein at least one of the second bit line and the second complementary bit line is disposed on the backside of the substrate below the substrate layer.
2. The semiconductor structure of claim 1, wherein the data storage element is a static random-access memory (SRAM) device.
3. The semiconductor structure of claim 1, wherein the first bit line and the second bit line are disposed on the front side of the substrate above the second active layer, and the first complementary bit line and the second complementary bit line are disposed on the backside of the substrate below the substrate layer.
4. The semiconductor structure of claim 1, wherein the first bit line and the first complementary bit line are disposed on the front side of the substrate above the second active layer, and the second bit line and the second complementary bit line are disposed on the backside of the substrate below the substrate layer.
5. The semiconductor structure of claim 1, wherein the first bit line set further includes a third bit line coupled to the first bit line, and the third bit line is disposed on a side of the substrate opposite to the first bit line.
6. The semiconductor structure of claim 1, wherein the second bit line set further includes a fourth bit line coupled to the second bit line, and the fourth bit line is disposed on a side of the substrate opposite to the second bit line.
7. The semiconductor structure of claim 1, further comprising: a power/ground line, disposed on the backside of the substrate below the substrate layer at an elevation same as that of the at least one of the second bit line and the second complementary bit line.
8. The semiconductor structure of claim 7, wherein the first bit line and the second bit line are disposed on the backside of the substrate below the substrate layer, and the power/ground line is disposed between the first bit line and the second bit line.
9. The semiconductor structure of claim 1, wherein the data storage element includes a transistor, and the power/ground line is connected to a source of the transistor from the backside of the substrate, and the at least one of the second bit line or the second complementary bit line is connected to a drain of the transistor from the backside of the substrate.
10. The semiconductor structure of claim 1, wherein the data storage element includes a transistor having a source, a gate and a drain arranged along a first direction, and the first bit line or the first complementary bit line extends substantially along the first direction.
11. A semiconductor structure, comprising: a substrate, having a front side and a backside opposite to the front side, wherein the substrate includes a substrate layer, a first active layer disposed above the substrate layer, and a second active layer disposed above the first active layer, the second active layer is proximal to the front side, and the substrate layer is proximal to the backside; a data storage element, disposed in and surrounding the first active layer and the second active layer; and a first

word line, disposed on the backside of the substrate below the substrate layer.

12. The semiconductor structure of claim 11, further comprising: a power/ground line, disposed on the backside of the substrate below the substrate layer, wherein the power/ground line is disposed at an elevation above an elevation of the first word line.

13. The semiconductor structure of claim 12, wherein an extending direction of the power/ground line is substantially perpendicular to an extending direction of the first word line.

14. The semiconductor structure of claim 11, further comprising: a second word line, disposed on the front side of the substrate above the second active layer.

15. The semiconductor structure of claim 14, further comprising: a third word line, disposed on the front side of the substrate at an elevation higher than an elevation of the first word line.

16. A semiconductor structure, comprising: a substrate, having a front side and a backside opposite to the front side, wherein the substrate includes a substrate layer proximal to the backside; a plurality of complementary FETs (CFETs), disposed on the substrate layer, wherein the CFETs are proximal to the front side of the substrate and collectively correspond to a data storage element; a first bit line set, including a first bit line and a first complementary bit line, wherein at least one of the first bit line and the first complementary bit line is disposed on the front side of the substrate above the CFETs; a second bit line set, including a second bit line and a second complementary bit line, wherein at least one of the second bit line and the second complementary bit line is disposed on the backside of the substrate below the substrate layer; and a first power/ground line, disposed on the backside of the substrate below the substrate layer at an elevation same as an elevation of the at least one of the second bit line and the second complementary bit line.

17. The semiconductor structure of claim 16, further comprising: a second power/ground line, disposed on the front side of the substrate above the CFETs at an elevation same as an elevation of the at least one of the first bit line and the first complementary bit line.

18. The semiconductor structure of claim 16, further comprising: a first word line, disposed on the backside of the substrate below the substrate layer, wherein the first word line is at an elevation below the elevation of the first power/ground line.

19. The semiconductor structure of claim 16, further comprising: a second word line, disposed on the front side of the substrate above the at least one of the first bit line and the first complementary bit line.

20. The semiconductor structure of claim 16, wherein the at least one of the second bit line and the second complementary bit line disposed on the backside of the substrate includes a first metallic material different from a second metallic material of the at least one of the first bit line and the first complementary bit line disposed on the front side of the substrate, and a resistance of the first metallic material is substantially less than a resistance of the second metallic material.
